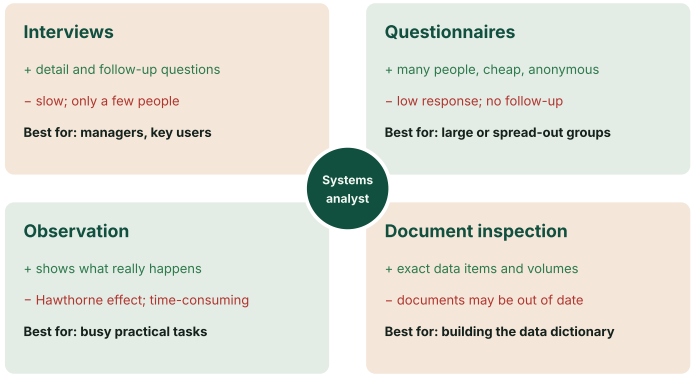
Analysis and fact-finding methods

AP03 FACT FILE

Specification: students should be able to

- Describe the purpose of the analysis (investigation) stage of systems development
- Explain the difference between direct and indirect data sources and state a benefit and drawback of each
- Describe the four main fact-finding techniques: interviews, questionnaires, observation and document sampling
- Evaluate the strengths and weaknesses of each fact-finding technique
- Select and justify suitable fact-finding techniques for a given scenario

Key words: analysis, systems analyst, fact-finding, direct data source, indirect data source, interview, structured interview, unstructured interview, questionnaire, closed question, open question, observation, Hawthorne effect, document sampling, triangulation, representative sample, quantitative data, qualitative data



The four fact-finding methods with an advantage, a disadvantage and the users each suits, around the systems analyst

Why analysis matters

Analysis, sometimes called the investigation stage, is where a systems analyst finds out exactly how an organisation's existing system works and what the new system must do. The analyst needs to understand the current system in terms of its inputs, the processing carried out on them, the outputs produced and the data that is stored. Just as importantly, the analyst identifies the strengths of the current system, which should be kept, its weaknesses and problems, which the new system must solve, and opportunities to add useful new features. Everything learned is used to agree the user requirements with the client before any design work begins. This matters because an error in understanding at this point is copied into the design, the code and the testing, and it becomes far more expensive to put right later. Because nobody can learn everything about an organisation from one conversation, the analyst gathers evidence using a range of fact-finding techniques: interviews, questionnaires, observation and document sampling. The findings are written up in a system specification that the client approves. A well-run analysis stage therefore reduces the risk of building a system that works technically but does not do what the users actually need.

Direct and indirect data sources

Data used to investigate a system or a business decision can come from direct or indirect sources. A direct data source is one created specifically for the current purpose, so the analyst collects the data first-hand without relying on anyone else, for example by writing and distributing a survey. Its great benefit is that the questions can be targeted at exactly the information needed. Its drawback is cost: time must be spent deciding what to ask, designing the questions, collecting responses and analysing them. An indirect data source already exists and was collected for a different reason, but is reused for the current purpose. Examples include government population statistics, a council's licensing records or an organisation's own sales history. Because the data has already been gathered and paid for, it is cheap and usually available straight away. However, it was not designed for the new purpose, so it may not contain precisely the data required, may be out of date or may need filtering and processing to remove irrelevant records. Although CCEA does not name this distinction in the specification outcomes, it is a useful way of thinking about evidence and it appears in textbook questions.

Interviews

Interviews are valuable because the staff who use a system every day usually know more about how it really works than anyone else. They are used to confirm the analyst's understanding, to fill gaps in it and to explore in depth an area where the interviewee is an expert. Good interviews need careful planning. The analyst decides precisely what must be found out, chooses interviewees carefully because interviewing everyone is neither possible nor necessary, and tells each person why they were selected, what the interview is for, how long it will take and how the information will be used. Some staff may fear that a new system threatens their job, so the analyst must be sensitive and reassuring. Interviews may be structured, with fixed questions that make comparison easy, or unstructured, allowing follow-up questions when something interesting emerges. Many analysts prefer to take notes and paraphrase answers back rather than record, since recording can make people guarded. The main strengths are depth of detail and flexibility; the main weaknesses are the time involved, the small number of people covered and possible bias, so answers should be cross-checked against other evidence.

Questionnaires

A questionnaire puts the same set of questions to many people and is the most practical method when the users are numerous or spread across several locations. Most are now completed online, which means respondents can answer privately, anonymously and at a time that suits them, and the software can total and chart the responses automatically. Typical input types are radio buttons, where only one option can be selected; checkboxes, where several options may be ticked; sliders or rating scales for a value on a range; and free text boxes for comments. Closed questions produce quantitative data that is quick to analyse, while open questions give qualitative data that is richer but slower to process. A well-designed questionnaire can include consistency checks, such as asking for age and later for date of birth, so that careless or untruthful responses can be filtered out. The disadvantages are significant. Response rates are often low, the analyst cannot ask a follow-up question when an answer is unclear, and once the questionnaire has been issued a confusing or leading question cannot be fixed. Writing clear, unbiased questions therefore takes time and should be tested on a small group first.

Observation

Observation means watching the system being used in its real environment, often by shadowing a representative sample of staff through their working day. It allows the analyst to record each process, the data needed to carry it out, the data it produces and how long it takes. It is especially useful because people do not always work in the way the official procedures describe. Staff often develop their own shortcuts and workarounds, some of which are more efficient than the documented method and should be kept in the new system, while others reveal genuine problems. The main weakness is the Hawthorne effect: people who know they are being watched may change their behaviour, for example by taking more care or following rules they normally skip, so the analyst sees an unrepresentative picture. This can be reduced by observing as unobtrusively as possible, watching several people and observing over a longer period. Observation is also time-consuming, may miss tasks that happen rarely, such as end-of-year processing, and is harder to use where similar work is done in slightly different ways at several sites, because local practices at one site may not apply elsewhere.

Document sampling and combining methods

Document sampling, also called document inspection, involves examining a sample of the paperwork and files the current system uses and produces, such as order forms, invoices, receipts, reports and procedure manuals. It is often a sensible first step because it shows the analyst who the key users are, which data items are recorded and in what format, how data moves between departments, the volume of data handled and the outputs the organisation relies on. It may also show how the system came about and how it has evolved. It is objective and does not interrupt staff, but it can be slow, and documentation may be incomplete or out of date. It also shows what is recorded rather than how or why people use it. Because every technique has weaknesses, analysts combine them and use triangulation, checking a fact from one source against another. For instance, a manager may claim a form is rarely used, but a sample of files may show hundreds a month. Such contradictions are worth investigating, as they frequently point to the real problems the new system must solve. The choice of methods depends on the number and location of users, the depth required, the time and budget available and the sensitivity of the topic.

T Key terms from the mark schemes

Analysis
The stage of systems development in which the existing system is investigated and the requirements for its replacement are identified and agreed with the client.
`Credited: Investigation`

Fact-finding
Gathering information about a current system, using techniques such as interviews, questionnaires, observation and document sampling, to establish how it works and what is required.

Systems analyst
The person who investigates the current system, establishes the requirements and specifies the new system.

Direct data source
A source of data collected first-hand specifically for the current purpose, such as a survey designed for the project.
`Credited: Primary data source`

Indirect data source
Data that already exists and is reused for a purpose other than the one for which it was originally collected.
`Credited: Secondary data source`

Hawthorne effect
The tendency of people to change their behaviour because they know they are being observed, which can make observation unreliable.
`Credited: Users act up when watched`

Closed question
A question with a fixed set of possible answers, producing quantitative data that is easy to analyse.

Open question
A question that lets the respondent answer in their own words, giving richer qualitative data that is harder to analyse.

Document sampling
Examining a sample of existing forms, invoices, reports and records to identify the data items, formats, volumes and outputs of the current system.
`Credited: Document inspection`

Triangulation
Checking a finding from one source of evidence against another to confirm that it is accurate and consistent.
`Credited: Cross-checking`

Exam focus

Short questions ask you to state or name techniques, give one benefit and one drawback, or explain why fact-finding is needed. For the purpose of analysis, mark schemes credit understanding the existing system's inputs, processes and outputs, identifying its strengths and weaknesses, and finding opportunities for improvement. When describing a technique, give its main features as well as an evaluation point: for observation, mention shadowing a representative sample and identifying the processes and data involved, then name the Hawthorne effect as a drawback. Avoid one-word evaluations such as 'quick' or 'cheap' unless you say compared with what and why. Scenario questions worth three or more marks expect you to choose a technique, say who it would be used with and justify it using details from the scenario, such as the number of users, their location or the type of information needed. In a 6 to 8 mark discussion, cover several techniques, give strengths and weaknesses of each and reach a reasoned conclusion about the best combination. Common mistakes are confusing direct and indirect sources, describing a questionnaire as allowing follow-up questions, and forgetting that interviews are time-consuming and only reach a few people.

? Check yourself

1. Before designing a new ticketing system for a Derry theatre, the systems analyst spends three weeks investigating how the box office currently works. State three things the analyst is trying to achieve during this investigation. [3]
2. A sports shop in Bangor is deciding whether to stock a new range of hurling equipment. Using this example, explain the difference between a direct and an indirect data source. [2]
3. The owner of a café in Portrush is considering opening a second café in Portstewart. Identify one direct and one indirect data source she could use, and state one drawback of each. [4]

Answers: 1. Any three from: to understand the existing system's inputs, processes and outputs (1) / to identify strengths of the current system to keep (1) / to identify weaknesses or problems the new system must solve (1) / to identify opportunities for improvement or new features (1) / to establish and agree the user requirements (1) / to provide the information needed for the system specification (1) 2. A direct data source is collected first-hand specifically for the current purpose, e.g. a survey of the shop's customers about hurling equipment (1) / an indirect data source already exists, was gathered for some other reason and is reused here, e.g. local GAA club membership figures or the shop's past sales records (1) 3. Direct, e.g. a survey/questionnaire of people in Portstewart, interviews with local residents or a trial pop-up stall (1); drawback: expensive/time-consuming to design, carry out and analyse (1). Indirect, e.g. NISRA population statistics, council business listings, sales data from the Portrush café (1); drawback: collected for another purpose so may not contain exactly the data needed / may need filtering / may be out of date (1)